Student Attendance Tracker - script.js

```
// ---------- Data (Local Storage) ----------
let students  = JSON.parse(localStorage.getItem("sat_students")   || "[]");
let records   = JSON.parse(localStorage.getItem("sat_attendance") || "[]"); // {sid, date, status}
let threshold = Number(localStorage.getItem("sat_threshold")) || 75;
let editingId = null;

const $ = id => document.getElementById(id);
const today = new Date().toISOString().slice(0, 10);

function save() {
  localStorage.setItem("sat_students", JSON.stringify(students));
  localStorage.setItem("sat_attendance", JSON.stringify(records));
  localStorage.setItem("sat_threshold", threshold);
}
function showMsg(text, type) { $("msg").textContent = text; $("msg").className = "msg " + type; }
function esc(t) { const d = document.createElement("div"); d.textContent = t; return d.innerHTML; }

// ---------- Attendance percentage ----------
function percent(sid) {
  const list = records.filter(r => r.sid === sid);
  if (!list.length) return null;
  const attended = list.filter(r => r.status === "Present" || r.status === "Late").length;
  return Math.round((attended / list.length) * 100);
}
const statusOn = (sid, date) => (records.find(r => r.sid === sid && r.date === date) || {}).status || "Not
    Marked";
const isLow = sid => percent(sid) !== null && percent(sid) < threshold;

// ---------- Add / Edit / Delete students ----------
function saveStudent() {
  const name = $("stuName").value.trim(), id = $("stuId").value.trim(), dept = $("stuDept").value.trim();
  if (!name || !id || !dept) return alert("Please fill all student details.");
  if (students.some(s => s.id === id && s.id !== editingId)) return alert("Student ID already exists!");
  if (editingId) {
    const s = students.find(x => x.id === editingId);
    records.forEach(r => { if (r.sid === editingId) r.sid = id; });
    Object.assign(s, { name, id, dept });
  } else {
    students.push({ name, id, dept });
  }
  resetForm(); save(); render();
}
function editStudent(id) {
  const s = students.find(x => x.id === id);
  editingId = id;
  $("stuName").value = s.name; $("stuId").value = s.id; $("stuDept").value = s.dept;
  $("formTitle").textContent = "Edit Student";
  $("saveBtn").textContent = "Update Student";
  $("cancelBtn").classList.remove("hidden");
}
function resetForm() {
  editingId = null;
  ["stuName", "stuId", "stuDept"].forEach(i => $(i).value = "");
  $("formTitle").textContent = "Add Student";
  $("saveBtn").textContent = "Add Student";
  $("cancelBtn").classList.add("hidden");
}
function deleteStudent(id) {
  if (!confirm("Delete this student and all attendance records?")) return;
  students = students.filter(s => s.id !== id);
  records = records.filter(r => r.sid !== id);
  $("historyBox").classList.add("hidden");
  save(); render();
}

// ---------- Mark attendance (no duplicates) ----------
function markAttendance() {
  const sid = $("attStudent").value, date = $("attDate").value, status = $("attStatus").value;
  if (!sid || !date) return showMsg("Select a student and a date.", "err");
  if (records.some(r => r.sid === sid && r.date === date))
    return showMsg("Attendance already marked for this student on " + date + ".", "err");
  records.push({ sid, date, status });
  save(); render();
  showMsg("" + status + " marked successfully.", "ok");
}

// ---------- History ----------
function showHistory(sid) {
```

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  const s = students.find(x => x.id === sid);
  const list = records.filter(r => r.sid === sid).sort((a, b) => b.date.localeCompare(a.date));
  $("historyTitle").textContent = "Attendance History – " + s.name + " (" + (percent(sid) ?? 0) + "%)";
  $("historyList").innerHTML = list.length
    ? list.map(r => `<span class="chip">${r.date} – <span class="badge
        ${r.status}">${r.status}</span></span>`).join("")
    : "No attendance records yet.";
  $("historyBox").classList.remove("hidden");
}

// ---------- Render UI ----------
function render() {
  const date = $("attDate").value || today;
  const q = $("search").value.toLowerCase(), dept = $("deptFilter").value, st = $("statusFilter").value;

  // summary
  const present = students.filter(s => ["Present", "Late"].includes(statusOn(s.id, date))).length;
  const absent  = students.filter(s => statusOn(s.id, date) === "Absent").length;
  const attended = records.filter(r => r.status === "Present" || r.status === "Late").length;
  $("sumTotal").textContent = students.length;
  $("sumPresent").textContent = present;
  $("sumAbsent").textContent = absent;
  $("sumPercent").textContent = (records.length ? Math.round(attended / records.length * 100) : 0) + "%";

  // dropdowns
  const depts = [...new Set(students.map(s => s.dept))];
  $("deptFilter").innerHTML = '<option value="">All Departments</option>' +
    depts.map(d => `<option ${d === dept ? "selected" : ""}>${esc(d)}</option>`).join("");
  const chosen = $("attStudent").value;
  $("attStudent").innerHTML = '<option value="">-- Select Student --</option>' +
    students.map(s => `<option value="${esc(s.id)}" ${s.id === chosen ? "selected" : ""}>${esc(s.id)} -
        ${esc(s.name)}</option>`).join("");

  // filtered table
  const rows = students.filter(s => {
    const status = statusOn(s.id, date);
    return (s.name.toLowerCase().includes(q) || s.id.toLowerCase().includes(q)) &&
      (!dept || s.dept === dept) &&
      (!st || (st === "Low" ? isLow(s.id) : status === st));
  });
  $("tableBody").innerHTML = rows.length ? rows.map(s => {
    const p = percent(s.id), status = statusOn(s.id, date);
    return `<tr class="${isLow(s.id) ? "low" : ""}">
      <td>${esc(s.id)}</td><td>${esc(s.name)}</td><td>${esc(s.dept)}</td>
      <td><span class="badge ${status}">${status}</span></td>
      <td>${p === null ? "–" : p + "%"}${isLow(s.id) ? '<span class="low-tag">LOW</span>' : ""}</td>
      <td>
        <button class="btn small edit" onclick="editStudent('${esc(s.id)}')">Edit</button>
        <button class="btn small hist" onclick="showHistory('${esc(s.id)}')">History</button>
        <button class="btn small del" onclick="deleteStudent('${esc(s.id)}')">Delete</button>
      </td></tr>`;
  }).join("") : '<tr><td colspan="6">No students found.</td></tr>';
}

// ---------- Events ----------
$("saveBtn").onclick = saveStudent;
$("cancelBtn").onclick = resetForm;
$("markBtn").onclick = markAttendance;
$("attDate").value = today;
$("attDate").onchange = render;
$("threshold").value = threshold;
$("threshold").oninput = e => { threshold = Number(e.target.value) || 75; save(); render(); };
["search", "deptFilter", "statusFilter"].forEach(i => $(i).oninput = render);
render();
```